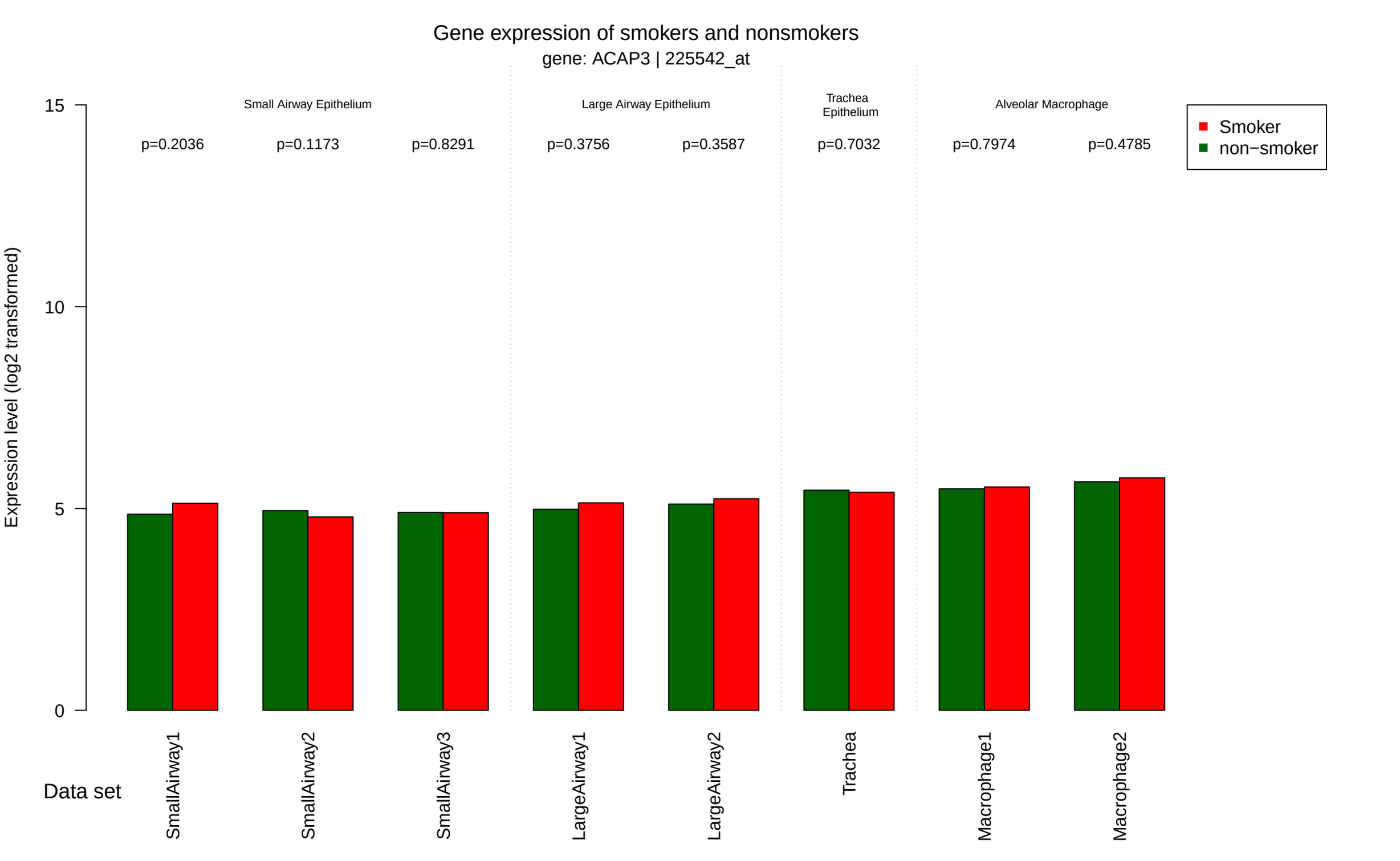
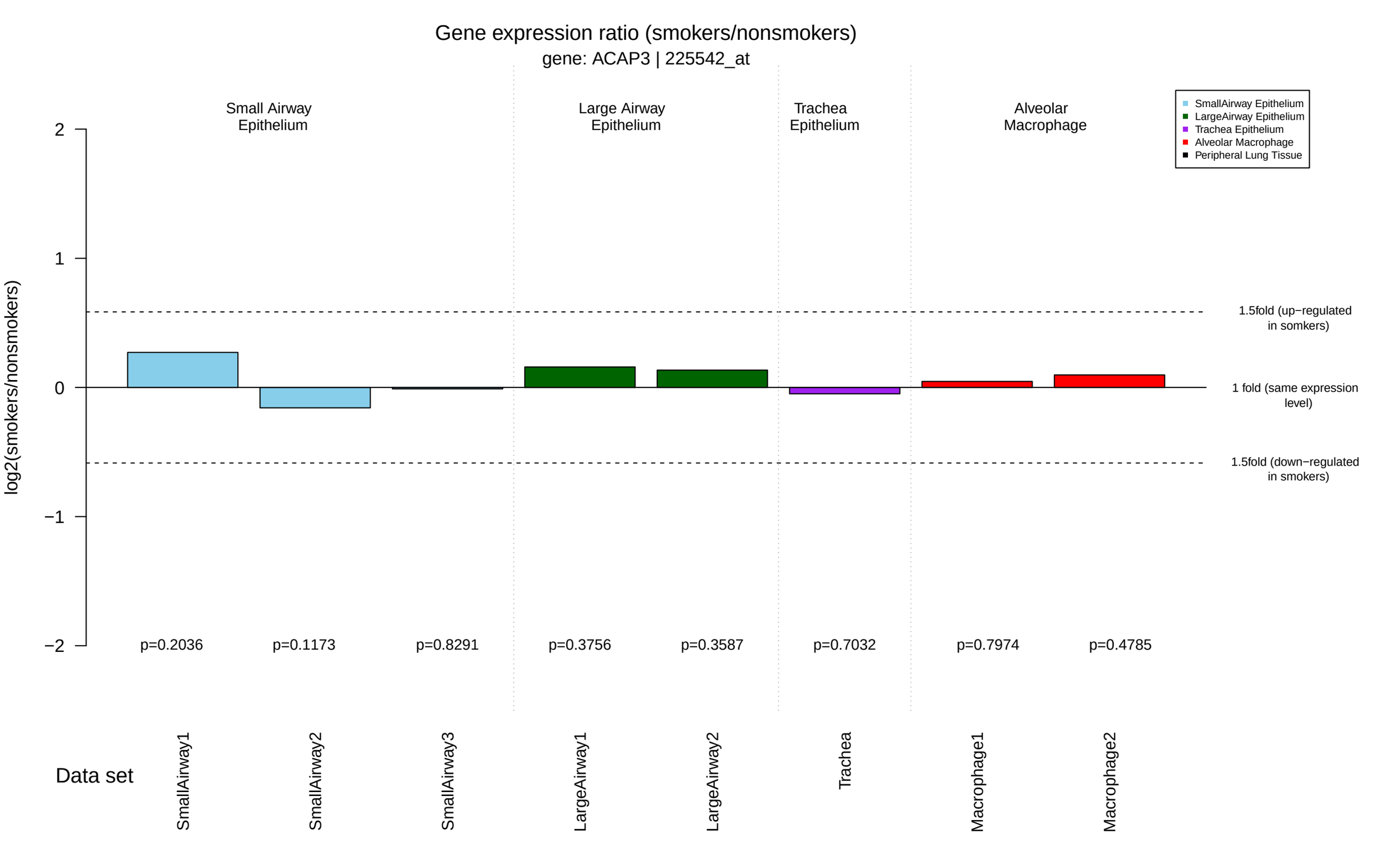
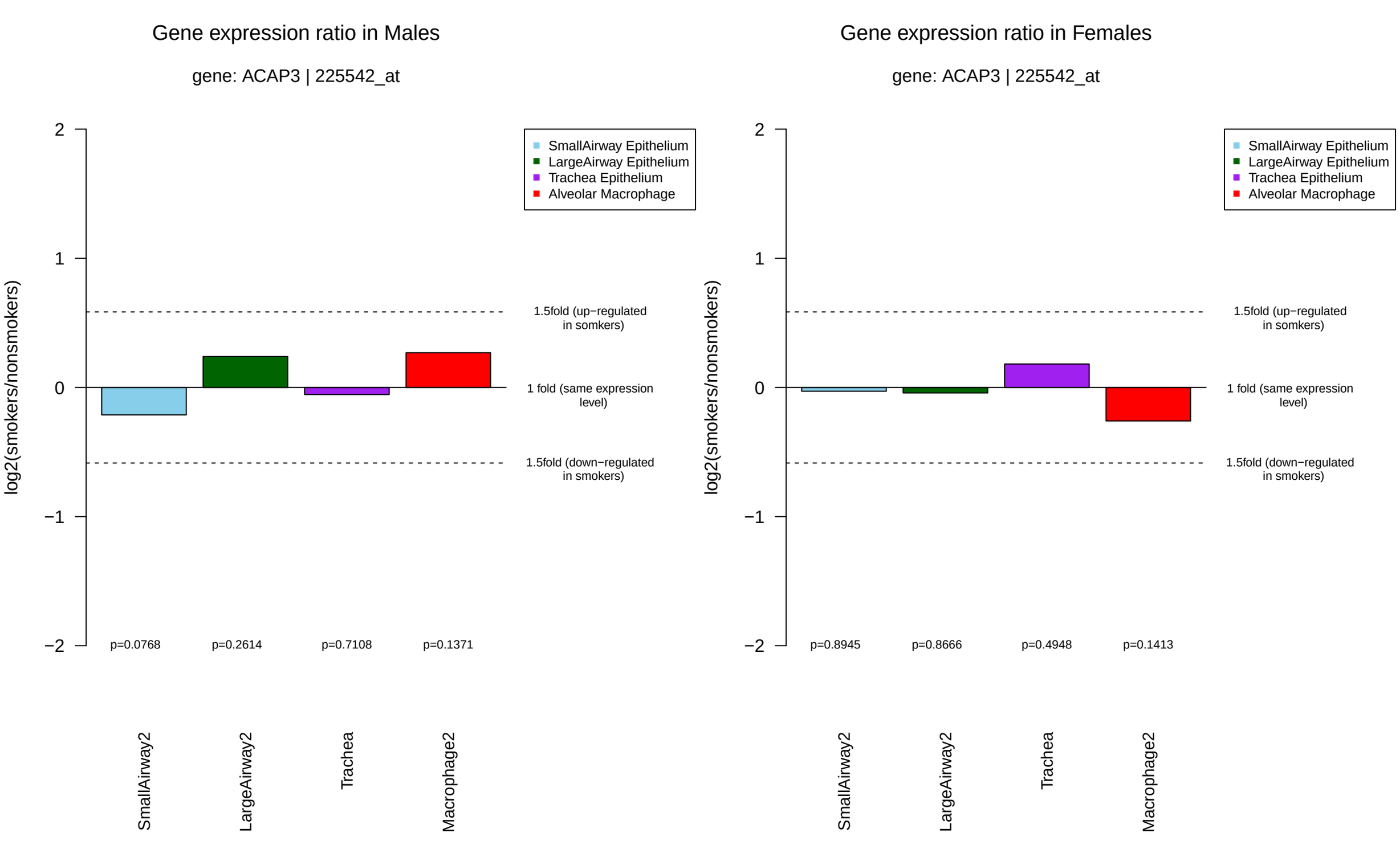
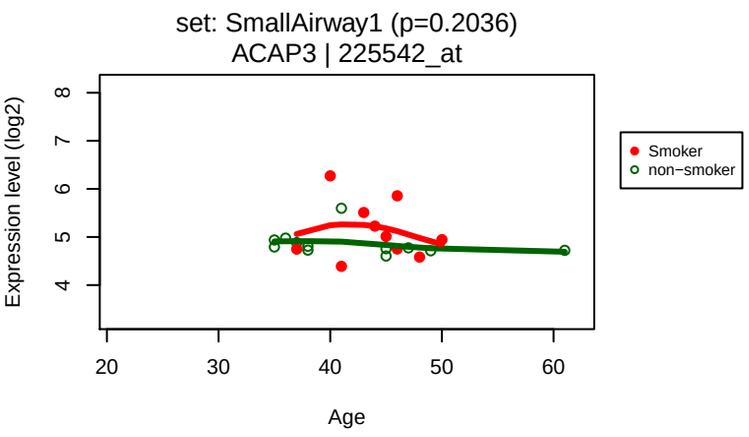
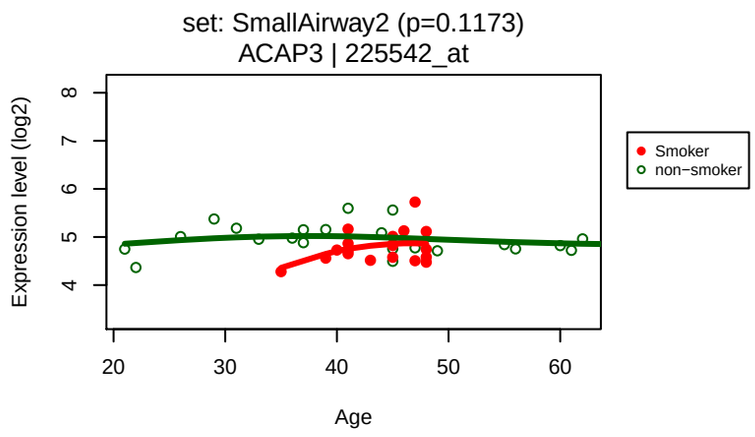
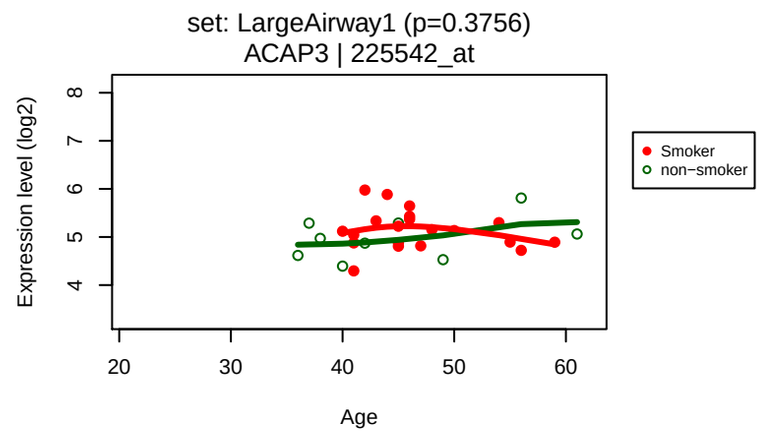

Gene expression of smokers and nonsmokers
gene: ACAP3 | 225542_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2036
p=0.1173
p=0.8291
p=0.3756
p=0.3587
p=0.7032
p=0.7974
p=0.4785
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

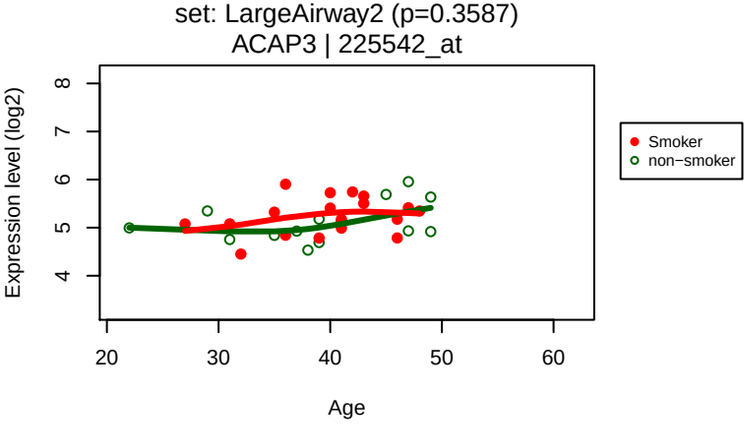

Gene expression ratio (smokers/nonsmokers)
gene: ACAP3 | 225542_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2036
p=0.1173
p=0.8291
p=0.3756
p=0.3587
p=0.7032
p=0.7974
p=0.4785
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

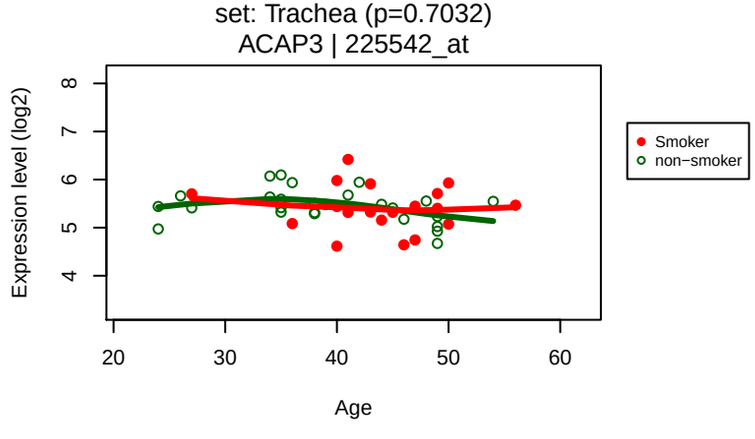

Gene expression ratio in Males
gene: ACAP3 | 225542_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0768
p=0.2614
p=0.7108
p=0.1371
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ACAP3 | 225542_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8945
p=0.8666
p=0.4948
p=0.1413
SmallAirway2
LargeAirway2
Trachea
Macrophage2

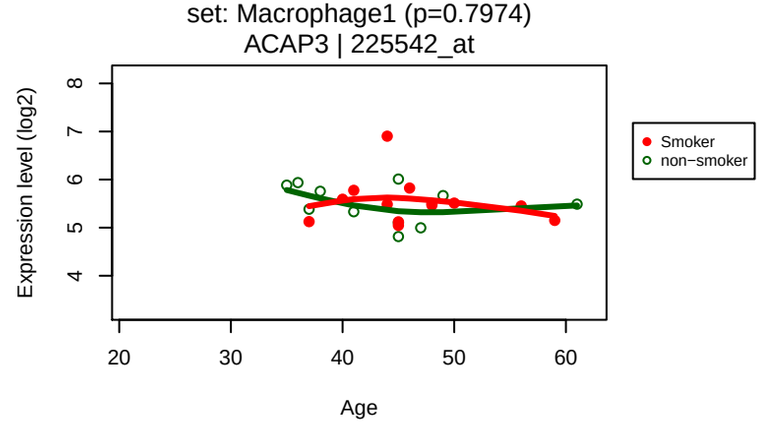

set: SmallAirway1 (p=0.2036)
ACAP3 | 225542_at
Expression level (log2)
Age
Smoker
non−smoker

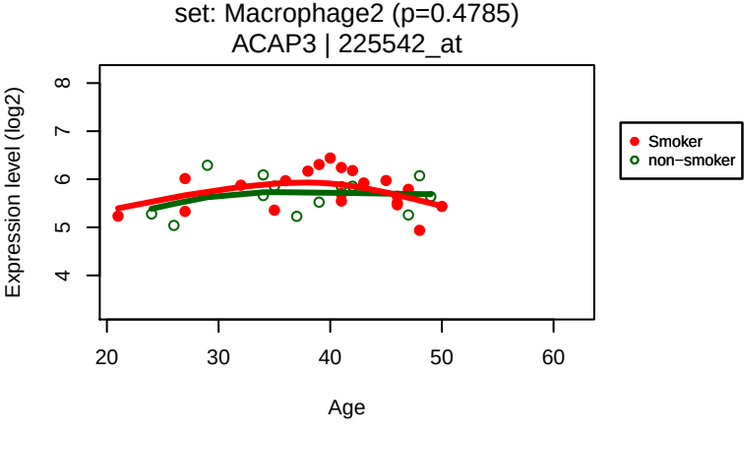

set: SmallAirway2 (p=0.1173)
ACAP3 | 225542_at
Expression level (log2)
Age
Smoker
non−smoker

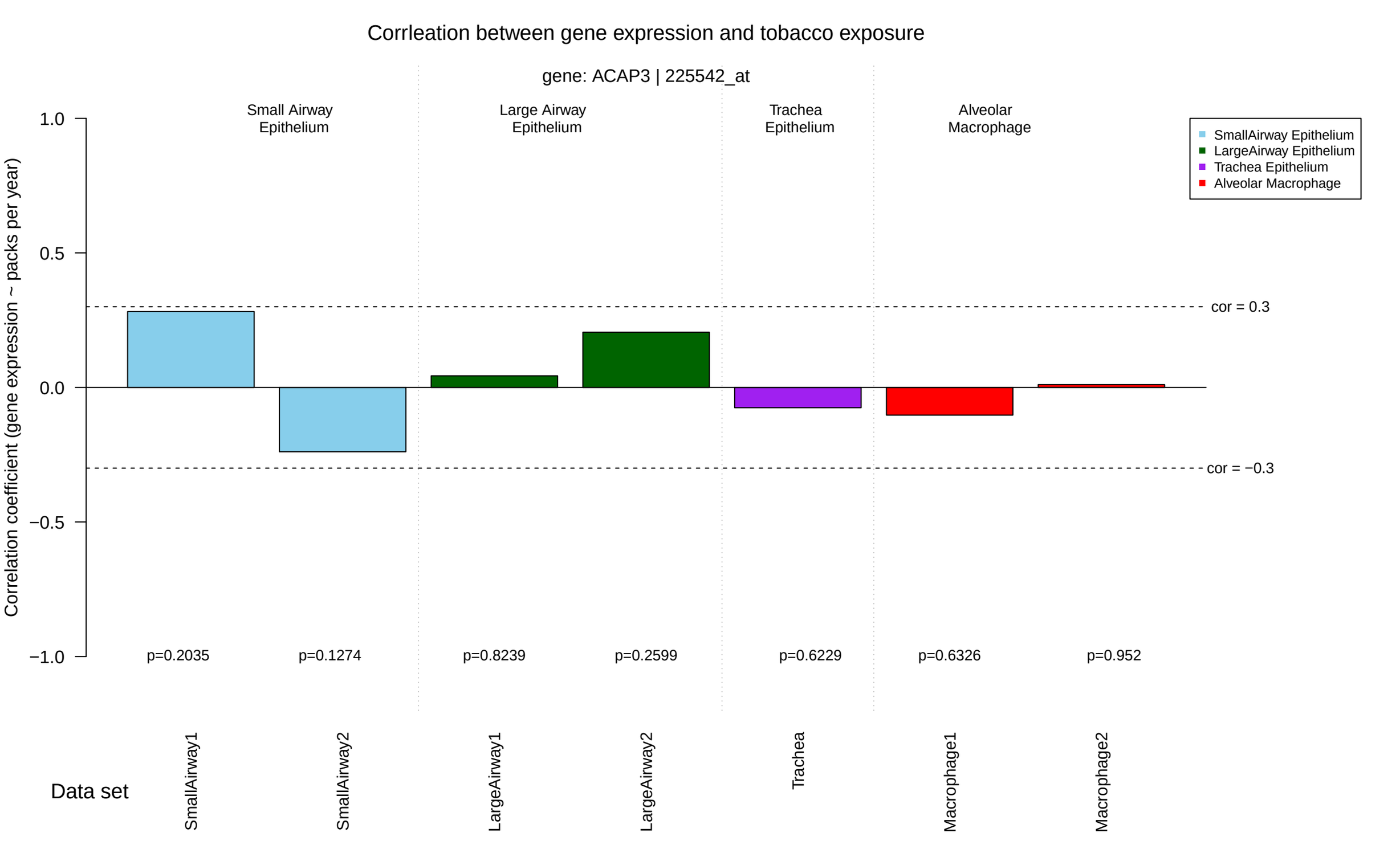

set: LargeAirway1 (p=0.3756)
ACAP3 | 225542_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3587)
ACAP3 | 225542_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7032)
ACAP3 | 225542_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7974)
ACAP3 | 225542_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4785)
ACAP3 | 225542_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ACAP3 | 225542_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2035
p=0.1274
p=0.8239
p=0.2599
p=0.6229
p=0.6326
p=0.952
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2